How Predictable Is NBA Cup Group Play? A Rules-Exact Simulator, a 2023–25 Backtest, and Two Studies of Early-Season Margins

Will Gunster · Sport Analytics, Syracuse University

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Abstract

The NBA Cup's group stage settles five-team groups in four games each, with tiebreaks that reward point differential, so a few points in one game can decide a group. This project builds a simulator that turns preseason win-total priors into team ratings on each team's real schedule and plays out every group under the official rules, which it reproduces exactly for 2023–25 (60 of 60 official placements, 16 of 16 seeds, 12 of 12 quarterfinal pairings). In a walk-forward backtest of the 18 groups of 2023–25, its group-winner forecasts tied with the pre-tournament market consensus, used only as a benchmark, with log loss 1.359 against 1.370, a difference of −0.011 per group (90% CI −0.053 to +0.030), with only 12% power to detect a gap of 0.02. Its skill over a uniform 1-in-5 guess is +15.5% (90% CI +3.9% to +27.2%). Two studies examine inputs a forecast rests on. In 1,506 October–November games, the pregame point spread moved about 0.8 points per point of missing player impact (95% CI 0.71 to 0.90), less than one-for-one, while actual margins moved 0.60 (95% CI 0.23 to 0.97). This is an association, not pre-registered. In an exploratory study of 10,899 games, teams on the second night of a back-to-back finished about 2.1 points below a strength-based expectation (95% CI about 1.5 to 2.6). A forecast of the 2026 Cup was frozen on October 14, 2026, before the first group game, and will be graded after group play ends on November 27.

1. Question

How predictable is NBA Cup group play, and what do absences and rest, two inputs a group forecast leans on, do to early-season margins? Four games is a short record. In a stylized calculation, chance alone gives a team's four-game record an SD of 0.25 against a talent SD of about 0.12 to 0.19, so 63–82% of the variance in group records is chance (an illustration, not data). The project asks whether a simulator built from public preseason information and the exact Cup rules forecasts group winners better than a uniform guess, how it compares with the pre-tournament market consensus, and how two game-level inputs, missing players and rest, relate to the spread and the margin.

2. Data

Input	Source	Coverage
Games, dates, venues, scores, overtime, Cup flags, pregame point spread	ESPN public scoreboard API	11,982 games, October 2016 to April 2026; 11,981 with a pregame spread
Who was listed and who played	ESPN game summaries (box scores)	1,716 games, October 1 to December 6 of 2021-22 to 2025-26
Official Cup standings and seeds	NBA Cup standings feed (2023, 2025); 2024 group tables citing nba.com	the oracle the tiebreak engine must reproduce
Team ratings	Basketball-Reference team summaries (SRS and margin of victory)	300 team-seasons, 2015-16 to 2025-26
Preseason priors	Preseason win totals (source recorded in the project's private provenance file)	one prior per team-season, 2016-17 to 2025-26, and October 14, 2026
Player ratings and projected minutes	DARKO eve-of-season tables (third-party)	name-free team totals only
Benchmark	The pre-tournament market consensus, used only as a benchmark	group-winner log loss per 2023–25 group

Rosters follow dated stints built from public season transaction lists (trades, waivers and signings). Player names and health information stay in a private data root, and the shared repository holds IDs and name-free team totals only.

3. Methods

3.1 The group simulator

Each team's preseason prior is converted to a point rating on its real schedule, with a strength scale of b = 0.906 (priors shrink toward the mean), strengths drawn around the ratings with SD tau 2.73 in each run, home court 2.51 points (0 at neutral sites) and a game-to-game SD of 12.61 points, all fit on 2,710 October–November games. Regulation margins are resampled from October–November residuals rather than drawn from a normal curve, and ties after regulation are modeled directly. Groups are then settled by the official tiebreaks: head-to-head record, then group point differential (an overtime game counts 0 from 2024 on), then total points, then the previous season's record. Gates run before any forecast or figure: the tiebreak engine must reproduce the official standings, the simulator must match exact enumeration, and a planted-

truth test must recover known parameters (calibration slope 1.000; log loss 1.280 against an oracle's 1.170).

3.2 The backtest

Each Cup of 2023–25 is forecast walk-forward, only with information available before it. The score is the log loss of the probability given to the eventual group winner: a uniform 1-in-5 guess scores ln 5 = 1.609, and skill is 1 minus the ratio of log loss to 1.609. The two hypotheses (skill above zero; model against the consensus) and the 90% interval rule were pre-registered. Intervals bootstrap the 18 groups.

3.3 The availability study

Missing impact for a team-game sums, over absent projected-rotation players (12+ projected minutes), the player's eve-of-season DARKO rating plus 1.5, times projected minutes over 48, and the regressor is home minus away. The outcomes are the pregame point spread (the home team's expected margin before tip) and the actual margin, each net of a preseason-strength expectation. Team-season fixed effects compare each team with itself on nights with and without absences. Intervals cluster by season (CR1, t with 4 df), and four other interval methods are reported beside them. The design was not pre-registered, so the result is labeled an association on observed games.

3.4 The rest study

The outcome is the home margin minus the two teams' full-season SRS gap. Four rest states enter as home-minus-away flags: the second night of a back-to-back, a third game in four nights, the first night of a back-to-back, and four or more days since the last game. The reference is a team with one or two days off. Intervals cluster by season (9 seasons, t with 8 df), and p-values are Holm-adjusted across the four states. The states were chosen after looking at the data, so the study is exploratory.

3.5 The 2026 forecast

The forecast runs 16 specifications of 1,000,000 simulations each: a central run, 13 one-at-a-time sensitivities and 2 joint scenarios. It was frozen on October 14, 2026, before any 2026 Cup game. Its SHA-256 is recorded in the repository, and the rebuild reproduces it bit for bit from the frozen name-free inputs.

4. Results

4.1 Group-winner forecasts, 2023–25 (backtest, pre-registered)

The model tied with the pre-tournament market consensus, at −0.011 per group for model minus consensus (90% CI −0.053 to +0.030; SE 0.026, 95% CI −0.061 to +0.038). The test had 12% power to detect a gap of 0.02, and about 236 groups (39 Cups) would be needed for 80%, but the 90% interval does rule out gaps larger than about 0.054 per group. The model scored better than prior-season

SRS used as declared (model minus SRS −0.218, 90% CI −0.387 to −0.059) and than SRS calibrated like the model, a version chosen after seeing the declared run (−0.148, −0.231 to −0.063). Its Brier skill is +12.7% (90% CI +0.7% to +24.7%; 95% CI −1.6% to +27.0%). The favorite won 8 of 18 groups (44.4%, 95% CI 24.6% to 66.3%). In a stylized simulation, even perfect knowledge of team strengths would make the favorite win only 53–58% of groups. With 18 groups the pre-registered calibration band cannot be tested (calibration slope 1.255, 90% CI 0.60 to 2.40).

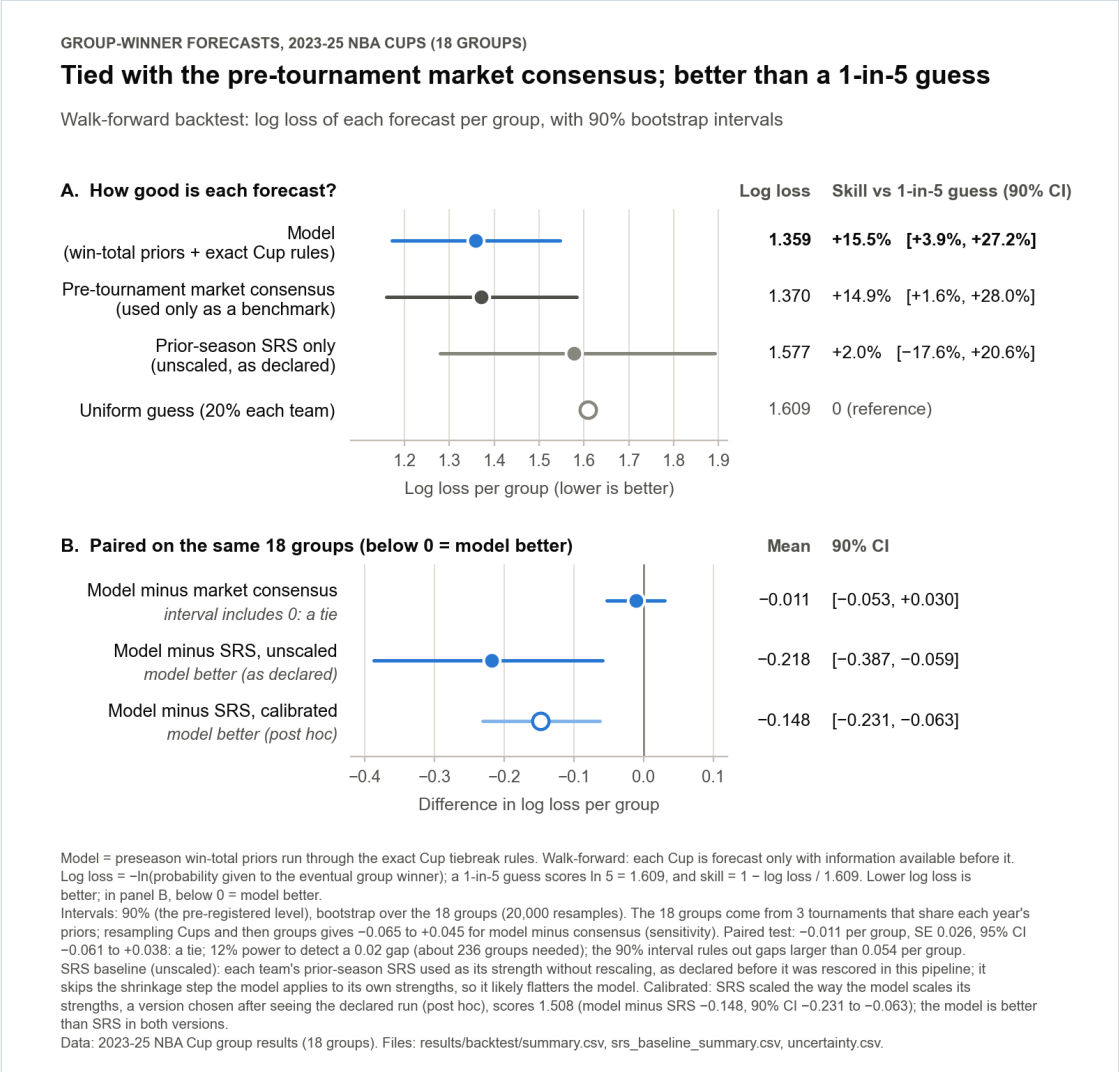


Figure 1. Group-winner forecasts in the 2023–25 walk-forward backtest. Panel A: log loss per group with 90% bootstrap intervals; panel B: paired differences on the same 18 groups (below 0 = model better).

4.2 How absences move the spread and the margin (association, not pre-registered)

The spread moves about 0.8 points per point of missing impact, less than one-for-one. Slope = 1 (the spread moves one-for-one with missing impact) is rejected, $p < 0.01$, and by the wild cluster bootstrap too (p about 0.02). Actual margins move 0.60 per point, and what the spread misses is not distinguishable from zero (−0.21, 95% CI −0.66 to 0.25). In decision units, a +2 DPM starter projected for 30 minutes is 2.19 impact points, so the spread moves about 1.8 points. Missing impact explains much of the spread (within R-squared 0.535) but little of any one game's margin

(0.013). The margin result rests on a few games. Without the 16 games with the largest missing-impact gap, the margin slope keeps its size (0.61) but its 95% interval runs from −0.03 to 1.25. Almost every team-game has someone out (98.1% of team-games had a projected-rotation player out, mean 3.35), and projected rotations cover 84.6% of October–November minutes. The slopes describe how the spread and the margin move with missing impact. They are not a value of any given player.

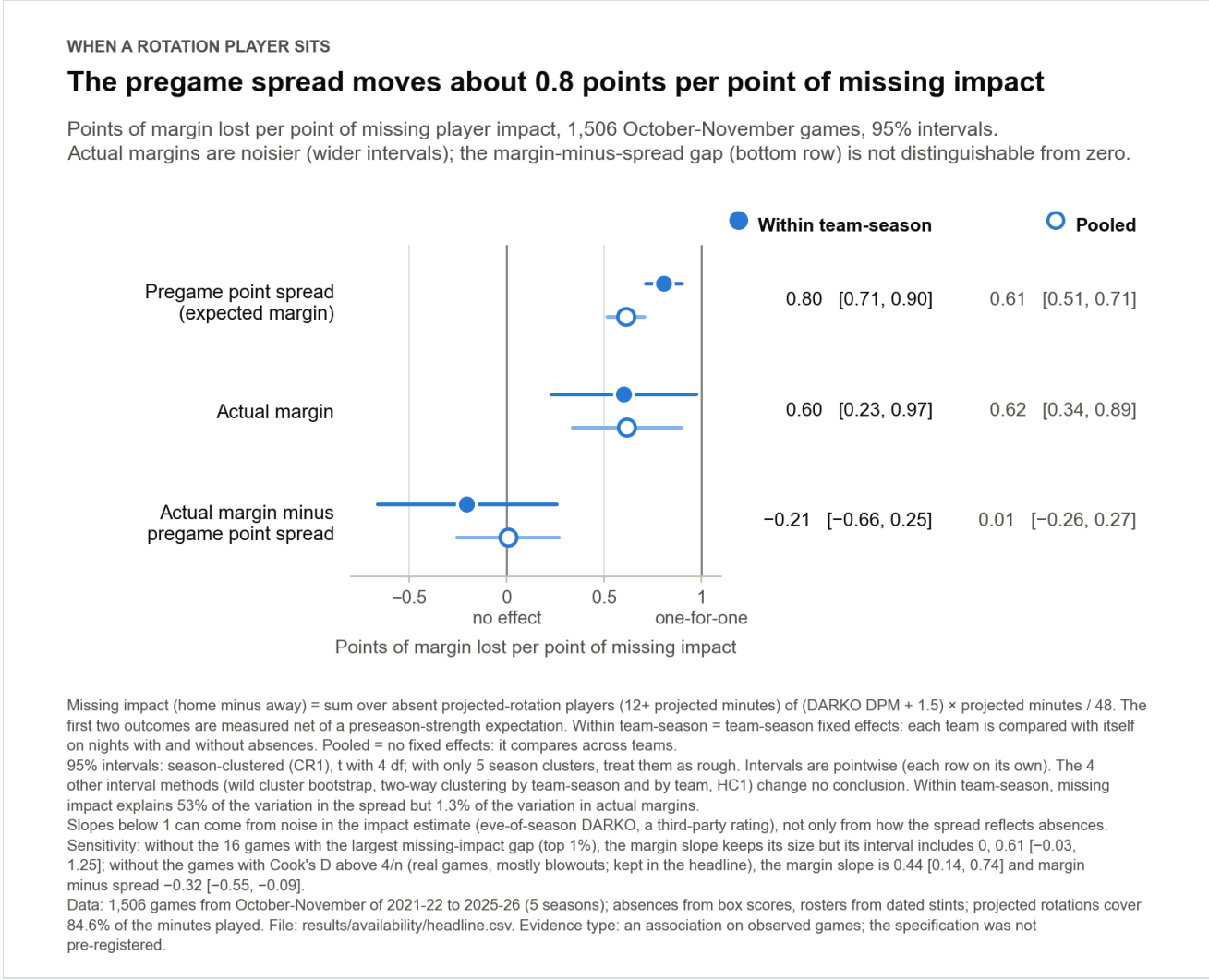


Figure 2. Points of margin lost per point of missing player impact, within team-season (filled) and pooled (hollow), with 95% season-clustered intervals; intervals are pointwise. Evidence type: an association on observed games.

4.3 Rest and back-to-backs (exploratory)

Teams on the second night of a back-to-back finished about 2.1 points below a strength-based expectation. The estimate holds under the other declared strength terms (−1.97 to −2.10) and is negative in all 9 seasons (−1.36 to −3.21). The third game in four nights, about 0.5 points, is unconfirmed after adjusting for the four tested states (raw p 0.040, Holm-adjusted p 0.12). Rest shifts the average but explains little of any one game, since the four flags explain about 0.9% of the

variation (R-squared 0.0089). The pregame spread already moves −1.90 for a second night (95% CI −2.07 to −1.74), and margin minus spread is −0.16 (−0.67 to +0.35). In an otherwise even game, 2.06 points is about 6.5 percentage points of win probability. A causal reading rests on one assumption, that the league sets the schedule before the season without giving stronger teams fewer back-to-backs.

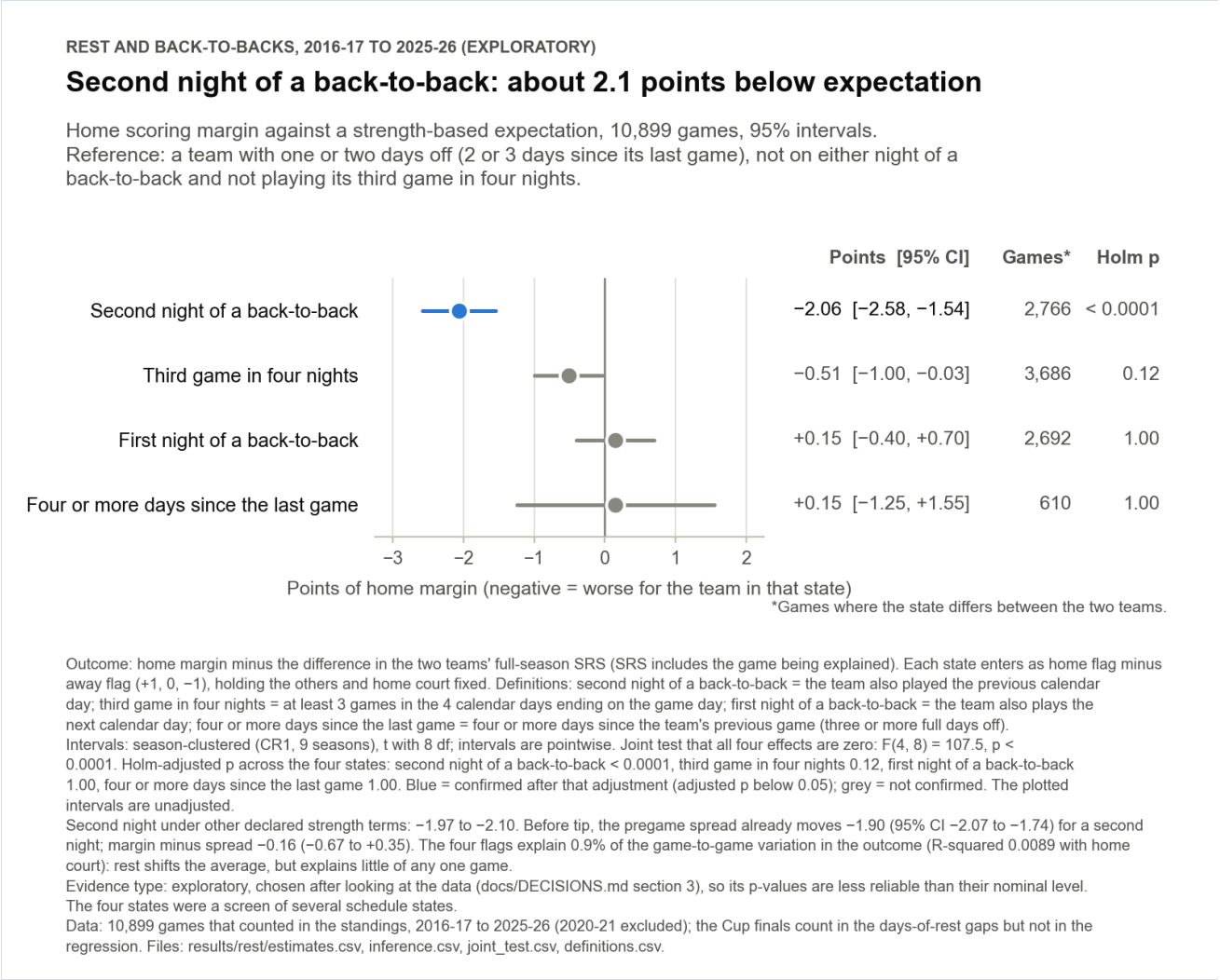


Figure 3. Rest states against a strength-based expectation of the home margin, with 95% season-clustered intervals and Holm-adjusted p-values; blue = confirmed after the adjustment, grey = not. Exploratory: the states were chosen after looking at the data.

4.4 Ties after regulation (exploratory)

633 of 11,981 games were tied after regulation (5.28%, 95% CI 4.9% to 5.7%), against 2.6% under a normal margin model, and every spread bin sits above the model. Why ties are this common was not tested. It matters for the Cup because an overtime game counts 0 in point differential from 2024 on, so the simulator models ties directly from resampled margins instead of from a normal curve.

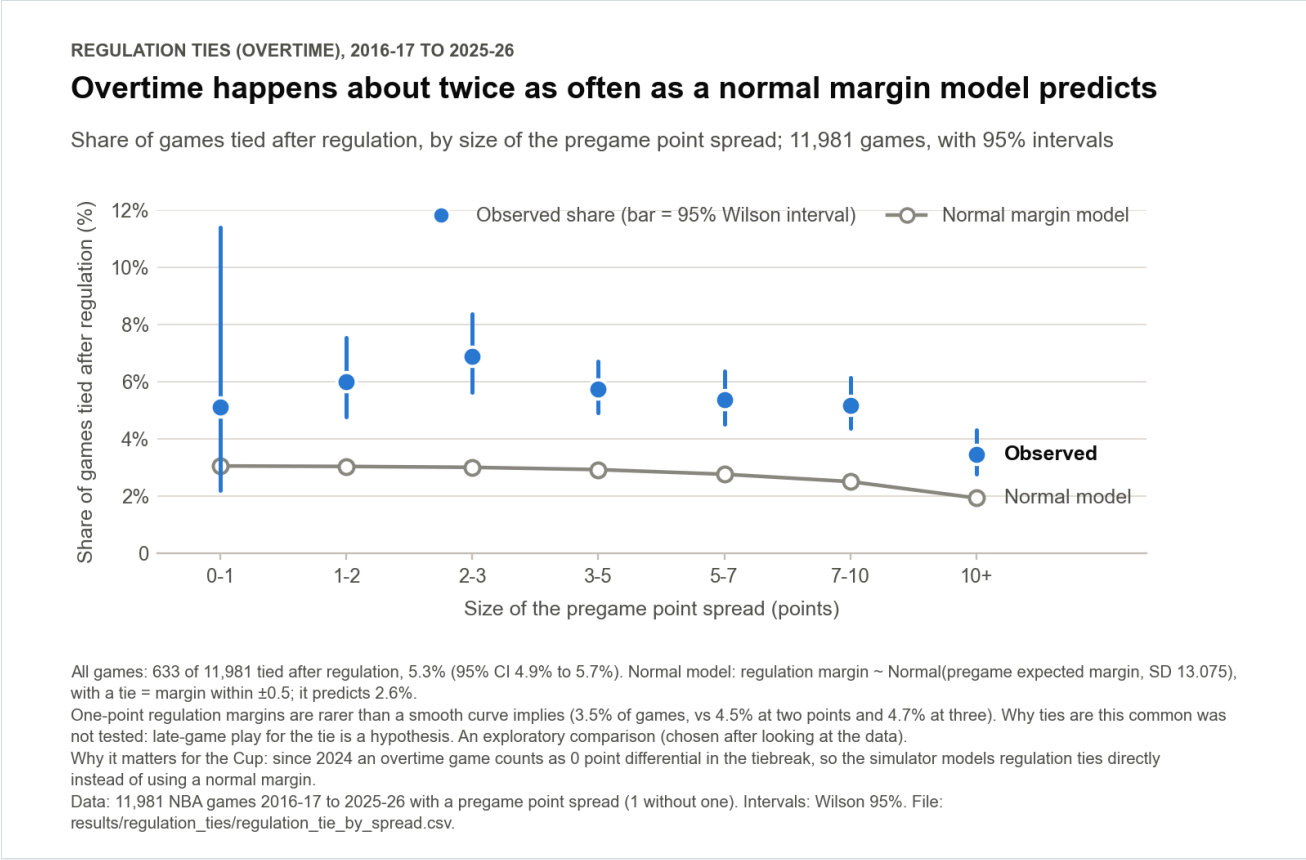


Figure 4. Share of games tied after regulation by the size of the pregame spread, observed against a normal margin model. Exploratory comparison; the mechanism was not tested.

4.5 The 2026 forecast (prospective, ungraded)

The frozen run gives Oklahoma City 56% and San Antonio 51% to win their groups, the only teams above 50%. East Group B is close to a five-way toss-up, with every team between 16% and 24%. The bars in Figure 5 are the range across the 16 specifications, not a confidence interval. Monte Carlo error is small next to them (the largest SE of a simulated probability is 0.0005 at 1,000,000 runs). The largest move of a group favorite under any specification is 12.3 points (Oklahoma City, under a joint scenario with b = 1 and a 33% Cup-effort term). The forecast will be graded after group play ends on November 27.

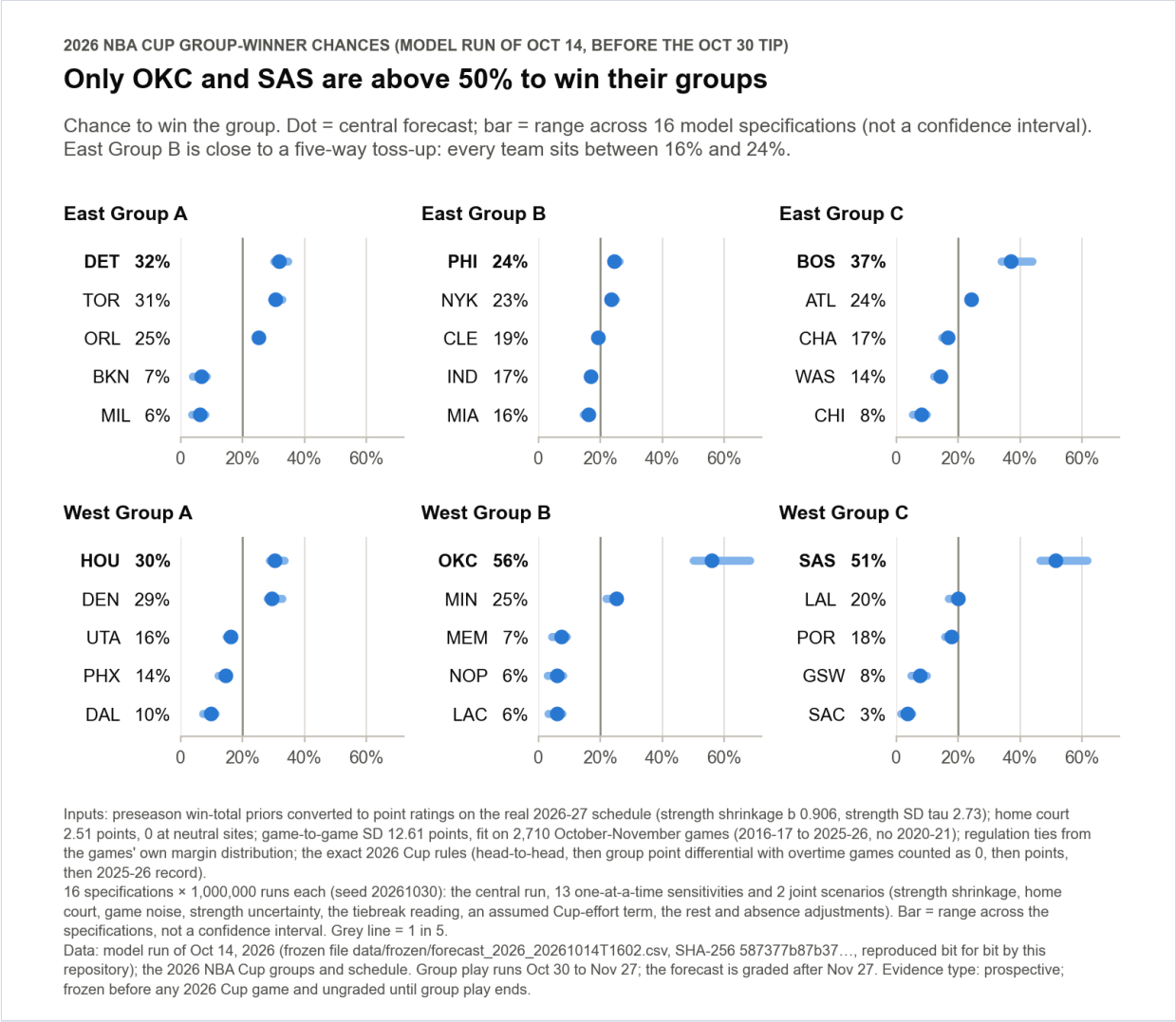


Figure 5. 2026 group-winner chances from the run frozen on October 14, 2026: dot = central forecast; bar = range across 16 specifications. Prospective; ungraded until group play ends.

5. Checks and diagnostics

- **Rules.** The engine reproduces the official 2023 and 2025 standings (60 of 60 placements, 16 of 16 seeds, 12 of 12 quarterfinal pairings) and the 2024 reference tables (points and differential 30 of 30, order 30 of 30, seeds 8 of 8, wild cards 2 of 2). 89 of 89 gate tests pass.
- **Leakage audit.** Removing information from after each forecast date moved the backtest skill from +16.0% to +15.5% and the gap to the benchmark from −0.018 to −0.011. The reported values are the leakage-free ones.
- **Structure.** Simulated groups look like real ones. The SD of teams' group point differential is 37.0 observed against a simulated mean of 32.6 (90% range 28.3 to 37.0), and the counts of 4-0 winners, first-place ties and group games tied after regulation also fall inside their simulated ranges.

- **Second route.** 112 of 112 second computations agree (tolerance 1e-8, counts exact), the oracle's 96 checks pass, and 142 tests pass.
- **Sharing.** A share check scans every shared file for names, paths and excluded terms before anything leaves the repository, with 0 hits.

6. Data corrections

A data validation on October 21–22, 2026 checked every dataset against independent sources. Roster coding was rebuilt from dated transactions: one trade's main player had been keyed to the wrong team, players stayed coded absent after a waiver or trade (12 player-team-seasons), and one player was coded absent in every game because two sources spelled his name differently, which an ID-level alias now fixes. The availability prior had been fit on seasons that included the season being analyzed, and it now uses earlier seasons only. ESPN had dated four February 2020 games a day late and given seven games the wrong arena, and rest gaps had been computed after dropping the Cup finals. Each fix was applied one at a time, with its effect on every headline metric recorded.

Quantity (95% CI)	As shipped	Corrected
Spread slope, within team-season	`0.81 (0.71 to 0.90)`	`0.80 (0.71 to 0.90)`
Margin slope, within team-season	`0.60 (0.21 to 0.99)`	`0.60 (0.23 to 0.97)`
Spread slope, pooled	`0.60 (0.54 to 0.66)`	`0.61 (0.51 to 0.71)`
Team-games with a rotation player out	`98.4%`	`98.1%`
Third game in four nights	`−0.52, "borderline"`	`−0.51, unconfirmed after Holm`
Prior-season SRS baseline (log loss; model minus SRS)	`1.506; −0.147`	`1.577; −0.218 (as declared)`

Table 1. The planted-truth test was also rerun with fitted parameters (the shipped run had fit on an empty template), and the claim that early-season margins are less noisy was dropped because robust measures disagree.

7. Limitations

Limit	Mechanism	Likely direction
Teams choose absences	Rest on back-to-backs and sitting players in lopsided games.	Understates a player's effect on the margin; the slope is not a value of a player.
Noisy impact measure	DARKO is an eve-of-season proxy, and its history replays today's model on past dates.	Attenuates both slopes, so "slope = 1 rejected" may partly reflect that noise.
Influential games	The margin result rests on 16 high-gap games.	Overstates how precisely the margin slope is known.
Rest: a bad control	Full-season SRS contains the game being explained.	Pulls the rest estimate toward zero.
Rest: travel	Travel and time zones are not measured.	Likely overstates the second-night effect if back-to-backs come with long trips.
Rest: states chosen after looking	Several schedule states were screened.	The printed p-values, even after Holm, are optimistic.
Few Cups	18 groups from 3 Cups share each year's priors.	The group bootstrap likely understates the uncertainty (resampling Cups, then groups: −0.065 to +0.045 for model minus consensus).
Cup incentives	Point-differential tiebreaks and overtime rules are not modeled as incentives.	The simulator likely understates the spread of group point differentials.

8. How a team could use it

- **Predictive.** The group simulator as an input or benchmark for Cup planning, for example locking in real results before the final group night and showing the margin each team needs to advance.
- **Predictive.** When a pregame spread is available, it already carries most of what public absence information says about the margin, so a model that adds absences on top of the spread risks counting them twice.
- **Causal, under the schedule assumption.** On the second night of a back-to-back, margins sit about 2.1 points below a strength-based expectation, about 6.5 percentage points of win probability in an otherwise even game, against about 7.9 for home court.
- **Not a player value.** The availability slopes describe how the spread and the margin move with missing impact. They are not a value of any given player.

9. Reproducibility and how this was built

One command, `make rebuild`, empties the results and figures folders and rebuilds everything in about 20 minutes, after an environment check that stops before anything is deleted if the Python version or a pinned package is wrong. A separate public rebuild runs without the private inputs. The gates run before any forecast or figure, every output is written atomically, and every number in the documents is rendered from the results rather than typed. The frozen 2026 forecast is never rerun or rewritten, and its hash is checked on every build.

The author directed the project and the October revision, which applied lessons from Syracuse coursework in statistics, sport analytics, finance and information studies: hypotheses and an interval rule fixed before the backtest, intervals clustered by season with alternative methods beside them, multiplicity adjustments across tested states, R-squared reported apart from significance, evidence-type labels on every number, and IDs instead of names in every join. AI coding tools (Claude Code) wrote the code and drafted this paper from the project's results files, and the October data validation and revision were run by AI agents under his direction.

Data sources (retrieval dates)

- ESPN public scoreboard API, 2016-10 to 2026-04 and the 2026-27 schedule: October 8, 2026. ESPN game summaries (box scores), 2021-22 to 2025-26: October 13, 2026.
- NBA Cup official standings feed (captures of December 3, 2023 and February 16, 2026): October 8, 2026. Wikipedia, 2024 NBA Cup group tables (citing nba.com): October 21, 2026.
- Basketball-Reference team summaries, 2015-16 to 2025-26: October 8, 2026. DARKO eve-of-season and season tables: October 8, 2026.
- Preseason win totals: October 8, 2026 (history) and October 14, 2026 (2026); source recorded in the private provenance file.
- NBA official full-season schedule (data.nba.com) and season transaction lists, for the date, arena and roster corrections: October 21, 2026.